Numa

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October 29, 2026

Repository: `https://github.com/DavidTilman/plpdi-mini-project`
Commit: `TODO`

1 Informal Description

2 Formal Syntax and Semantics

2.1 Abstract Syntax

$\oplus$	::=	$+ \mid - \mid * \mid / \mid \% \mid < \mid <= \mid == \mid != \mid >= \mid > \mid \mid\mid \mid \&\&$	binary operators
τ	::=	$\texttt{Nat} \mid \texttt{Int} \mid \texttt{Real} \mid \texttt{Bool} \mid (\tau_1, \ldots, \tau_n) \to \tau$	types
e	::=	$n \mid \texttt{True} \mid \texttt{False} \mid x \mid \texttt{Lambda}\ (x_1 : \tau_1, \ldots, x_n : \tau_n) : \tau \Rightarrow e$	literals, variable, lambda
	$\mid$	$-e \mid e_1 \oplus e_2 \mid e(e_1, \ldots, e_n) \mid e\ ?\ e_1 : e_2$	operations, application, ternary
s	::=	$\texttt{Var}\ x : \tau = e \mid \texttt{Var}\ x : \tau = \texttt{Read} \mid x := e$	declaration, read, assign
	$\mid$	$\texttt{While}\ (e)\ b \mid \texttt{If}\ (e)\ b_1\ \texttt{Else}\ b_2$	control flow
	$\mid$	$\texttt{Ret}\ e \mid \texttt{Print}\ e$	return or print
d	::=	$\texttt{Func}\ F\ (x_1 : \tau_1, \ldots, x_n : \tau_n) : \tau = b$	function declaration
b	::=	$\{\ s_1;\ \ldots;\ s_n;\ \}$	block ($n \geq 0$)
c	::=	$d \mid s$	top-level item
p	::=	$c_1;\ \ldots;\ c_n;$	program

2.2 Type System

The type system uses three partial meta-functions, writing $\text{Num} = \{\texttt{Nat}, \texttt{Int}, \texttt{Real}\}$ for the numeric types: neg types unary negation (widening a `Nat` to `Int`), widest is the least upper bound of $\sqsubseteq$ (used by the ternary), and sig gives the result type of a binary operator.

$$\mathsf{neg}(\tau) = \begin{cases} \texttt{Int} & \tau = \texttt{Nat} \\ \texttt{Int} & \tau = \texttt{Int} \\ \texttt{Real} & \tau = \texttt{Real} \\ \text{undefined} & \text{otherwise} \end{cases} \qquad \mathsf{widest}(\tau_1, \tau_2) = \begin{cases} \tau_1 & \tau_1 = \tau_2 \\ \text{the wider of } \tau_1, \tau_2 & \tau_1, \tau_2 \in \{\texttt{Nat}, \texttt{Int}, \texttt{Real}\} \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$\mathsf{sig}(\oplus, \tau_1, \tau_2) = \begin{cases} \mathsf{widest}(\tau_1, \tau_2) & \oplus \in \{+, *, /, \%\},\ \tau_1, \tau_2 \in \text{Num} \\ \mathsf{widest}(\mathsf{widest}(\tau_1, \tau_2), \texttt{Int}) & \oplus = -,\ \tau_1, \tau_2 \in \text{Num} \\ \texttt{Bool} & \oplus \in \{<, <=, >=, >\},\ \tau_1, \tau_2 \in \text{Num} \\ \texttt{Bool} & \oplus \in \{==, !=\},\ \tau_1, \tau_2 \in \text{Num} \text{ or } \tau_1 = \tau_2 = \texttt{Bool} \\ \texttt{Bool} & \oplus \in \{\&\&, ||\},\ \tau_1 = \tau_2 = \texttt{Bool} \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$\frac{}{t \sqsubseteq t}\ \text{S-REF} \qquad \frac{t \sqsubseteq t' \qquad t' \sqsubseteq t''}{t \sqsubseteq t''}\ \text{S-TRANS} \qquad \frac{}{\texttt{Nat} \sqsubseteq \texttt{Int}}\ \text{S-NatInt} \qquad \frac{}{\texttt{Int} \sqsubseteq \texttt{Real}}\ \text{S-IntReal}$$

$$\frac{}{\Gamma \vdash n : \texttt{Nat}}\ \text{TE-NAT} \qquad \frac{}{\Gamma \vdash n : \texttt{Real}}\ \text{TE-REAL} \qquad \frac{}{\Gamma \vdash \texttt{True} : \texttt{Bool}}\ \text{TE-TRUE}$$

$$\frac{}{\Gamma \vdash \texttt{False} : \texttt{Bool}}\ \text{TE-FALSE}$$

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau}\ \text{TE-VAR} \qquad \frac{\Gamma, x_1 : \tau_1, \ldots, x_n : \tau_n \vdash e : \tau' \qquad \tau' \sqsubseteq \tau}{\Gamma \vdash \texttt{Lambda}\,(x_1 : \tau_1, \ldots, x_n : \tau_n) : \tau \Rightarrow e : (\tau_1, \ldots, \tau_n) \to \tau}\ \text{TE-LAMBDA}$$

$$\frac{\Gamma \vdash e : \tau \qquad \mathsf{neg}(\tau) = \tau'}{\Gamma \vdash -e : \tau'}\ \text{TE-NEGATE} \qquad \frac{\Gamma \vdash e_1 : \tau_1 \qquad \Gamma \vdash e_2 : \tau_2 \qquad \mathsf{sig}(\oplus, \tau_1, \tau_2) = \tau}{\Gamma \vdash e_1 \oplus e_2 : \tau}\ \text{TE-OP}$$

$$\frac{\Gamma \vdash e : (\tau_1, \ldots, \tau_n) \to \tau \qquad \Gamma \vdash e_i : \tau_i' \qquad \tau_i' \sqsubseteq \tau_i \quad (1 \le i \le n)}{\Gamma \vdash e(e_1, \ldots, e_n) : \tau}\ \text{TE-APP}$$

$$\frac{\Gamma \vdash e : \texttt{Bool} \qquad \Gamma \vdash e_1 : \tau_1 \qquad \Gamma \vdash e_2 : \tau_2 \qquad \tau = \mathsf{widest}(\tau_1, \tau_2)}{\Gamma \vdash (e\ ?\ e_1\ :\ e_2) : \tau}\ \text{TE-COND}$$

$$\frac{\Gamma \vdash e : \tau' \qquad \tau' \sqsubseteq \tau}{\Gamma \vdash_{\tau_r} \texttt{Var}\ x : \tau = e \dashv \Gamma, x : \tau}\ \text{TS-DECL} \qquad \frac{\text{Read input is of type } \tau}{\Gamma \vdash_{\tau_r} \texttt{Var}\ x : \tau = \texttt{Read} \dashv \Gamma, x : \tau}\ \text{TS-READ}$$

$$\frac{x : \tau \in \Gamma \qquad \Gamma \vdash e : \tau' \qquad \tau' \sqsubseteq \tau}{\Gamma \vdash_{\tau_r} x := e \dashv \Gamma}\ \text{TS-ASSIGN}$$

$$\frac{\Gamma \vdash e : \texttt{Bool} \qquad \Gamma \vdash_{\tau_r} b}{\Gamma \vdash_{\tau_r} \texttt{While}\ (e)\ b \dashv \Gamma}\ \text{TS-WHILE} \qquad \frac{\Gamma \vdash e : \texttt{Bool} \qquad \Gamma \vdash_{\tau_r} b_1 \qquad \Gamma \vdash_{\tau_r} b_2}{\Gamma \vdash_{\tau_r} \texttt{If}\ (e)\ b_1\ \texttt{Else}\ b_2 \dashv \Gamma}\ \text{TS-COND}$$

$$\frac{\Gamma \vdash e : \tau' \qquad \tau' \sqsubseteq \tau_r}{\Gamma \vdash_{\tau_r} \texttt{Ret}\ e \dashv \Gamma}\ \text{TS-RET} \qquad \frac{\Gamma \vdash e : \tau}{\Gamma \vdash_{\tau_r} \texttt{Print}\ e \dashv \Gamma}\ \text{TS-PRINT}$$

$$\frac{\Gamma_0 \vdash_{\tau_r} s_1 \dashv \Gamma_1 \qquad \Gamma_1 \vdash_{\tau_r} s_2 \dashv \Gamma_2 \qquad \cdots \qquad \Gamma_{n-1} \vdash_{\tau_r} s_n \dashv \Gamma_n}{\Gamma_0 \vdash_{\tau_r} \{\, s_1; \ldots; s_n \,\}}\ \text{TB-BLOCK}$$

$$\frac{\begin{array}{c}\Gamma_0 = \overline{f_j : (\tau_{j,1}, \ldots, \tau_{j,k_j}) \to \tau_j'} \\ \left(\Gamma_0,\ x_{j,1} : \tau_{j,1}, \ldots, x_{j,k_j} : \tau_{j,k_j} \vdash_{\tau_j'} b_j\right)_j \qquad \Gamma_0 \vdash_\bullet \{\, s_1; \ldots; s_n \,\}\end{array}}{\vdash p \text{ is well-typed}}\ \text{TP-PROG}$$

2.3 Operational Semantics

$$\frac{}{\langle n, \rho, \sigma\rangle \Downarrow \langle n, \sigma\rangle}\ \text{E-Num} \qquad \frac{}{\langle \texttt{True}, \rho, \sigma\rangle \Downarrow \langle \texttt{True}, \sigma\rangle}\ \text{E-True} \qquad \frac{}{\langle \texttt{False}, \rho, \sigma\rangle \Downarrow \langle \texttt{False}, \sigma\rangle}\ \text{E-False}$$

$$\frac{}{\langle \texttt{Lambda}\ (\overline{x : \tau}) : \tau' \Rightarrow e, \rho, \sigma\rangle \Downarrow \langle \mathsf{clo}(\overline{x},\ \{\texttt{Ret}\ e;\},\ \rho), \sigma\rangle}\ \text{E-Lambda}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle}{\langle -e, \rho, \sigma\rangle \Downarrow \langle \hat{-} v, \sigma'\rangle}\ \text{E-Neg} \qquad \frac{\langle e_1, \rho, \sigma\rangle \Downarrow \langle v_1, \sigma_1\rangle \qquad \langle e_2, \rho, \sigma_1\rangle \Downarrow \langle v_2, \sigma_2\rangle}{\langle e_1 \oplus e_2, \rho, \sigma\rangle \Downarrow \langle v_1 \hat{\oplus} v_2, \sigma_2\rangle}\ \text{E-Op}$$

$$\frac{\begin{array}{c}\langle e, \rho, \sigma\rangle \Downarrow \langle \mathsf{clo}(\overline{x},\ b,\ \rho_\theta), \sigma_0\rangle \qquad \langle e_i, \rho, \sigma_{i-1}\rangle \Downarrow \langle v_i, \sigma_i\rangle \quad (1 \le i \le n) \\ \overline{\ell} \notin \mathrm{dom}(\sigma_n) \qquad \langle b, \rho_\theta[\overline{x} \mapsto \overline{\ell}], \sigma_n[\overline{\ell} \mapsto \overline{v}]\rangle \Downarrow \langle \mathsf{ret}\ v, \sigma'\rangle\end{array}}{\langle e(\overline{e}), \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle}\ \text{E-App}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{True}, \sigma'\rangle \qquad \langle e_1, \rho, \sigma'\rangle \Downarrow \langle v_1, \sigma''\rangle}{\langle e\ ?\ e_1 : e_2, \rho, \sigma\rangle \Downarrow \langle v_1, \sigma''\rangle}\ \text{E-Cond-T}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{False}, \sigma'\rangle \qquad \langle e_2, \rho, \sigma'\rangle \Downarrow \langle v_2, \sigma''\rangle}{\langle e\ ?\ e_1 : e_2, \rho, \sigma\rangle \Downarrow \langle v_2, \sigma''\rangle}\ \text{E-Cond-F}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle \qquad \ell \notin \mathrm{dom}(\sigma')}{\langle \texttt{Var}\ x : \tau = e, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho[x \mapsto \ell], \sigma'[\ell \mapsto v]\rangle}\ \text{S-Var}$$

$$\frac{v \text{ of type } \tau \text{ is read from input} \qquad \ell \notin \mathrm{dom}(\sigma)}{\langle \texttt{Var}\ x : \tau = \texttt{Read}, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho[x \mapsto \ell], \sigma[\ell \mapsto v]\rangle}\ \text{S-Read}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle \qquad \rho(x) = \ell}{\langle x := e, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho, \sigma'[\ell \mapsto v]\rangle}\ \text{S-Assign}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{True}, \sigma'\rangle \qquad \langle b, \rho, \sigma'\rangle \Downarrow \langle \mathsf{normal}, \sigma''\rangle \qquad \langle \texttt{While}\ (e)\ b, \rho, \sigma''\rangle \Downarrow \langle r, \rho, \sigma'''\rangle}{\langle \texttt{While}\ (e)\ b, \rho, \sigma\rangle \Downarrow \langle r, \rho, \sigma'''\rangle}\ \text{S-While-T}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{False}, \sigma'\rangle}{\langle \texttt{While}\ (e)\ b, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho, \sigma'\rangle}\ \text{S-While-F}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{True}, \sigma'\rangle \qquad \langle b, \rho, \sigma'\rangle \Downarrow \langle \mathsf{ret}\ v, \sigma''\rangle}{\langle \texttt{While}\ (e)\ b, \rho, \sigma\rangle \Downarrow \langle \mathsf{ret}\ v, \rho, \sigma''\rangle}\ \text{S-While-Ret}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{True}, \sigma'\rangle \qquad \langle b_1, \rho, \sigma'\rangle \Downarrow \langle r_1, \sigma''\rangle}{\langle \texttt{If}\ (e)\ b_1\ \texttt{Else}\ b_2, \rho, \sigma\rangle \Downarrow \langle r_1, \rho, \sigma''\rangle}\ \text{S-COND-T}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle \texttt{False}, \sigma'\rangle \qquad \langle b_2, \rho, \sigma'\rangle \Downarrow \langle r_2, \sigma''\rangle}{\langle \texttt{If}\ (e)\ b_1\ \texttt{Else}\ b_2, \rho, \sigma\rangle \Downarrow \langle r_2, \rho, \sigma''\rangle}\ \text{S-COND-F}$$

$$\frac{\langle e, \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle}{\langle \texttt{Ret}\ e, \rho, \sigma\rangle \Downarrow \langle \mathsf{ret}\ v, \rho, \sigma'\rangle}\ \text{S-Ret} \qquad \frac{\langle e, \rho, \sigma\rangle \Downarrow \langle v, \sigma'\rangle}{\langle \texttt{Print}\ e, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho, \sigma'\rangle}\ \text{S-Print}$$

$$\frac{}{\langle \{\ \}, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \sigma\rangle}\ \text{B-Empty} \qquad \frac{\langle s_1, \rho, \sigma\rangle \Downarrow \langle \mathsf{normal}, \rho', \sigma'\rangle \qquad \langle \{\ \overline{s}\ \}, \rho', \sigma'\rangle \Downarrow \langle r, \sigma''\rangle}{\langle \{s_1; \overline{s};\}, \rho, \sigma\rangle \Downarrow \langle r, \sigma''\rangle}\ \text{B-Seq}$$

$$\frac{\langle s_1, \rho, \sigma\rangle \Downarrow \langle \mathsf{ret}\ v, \rho', \sigma'\rangle}{\langle \{s_1; \overline{s};\}, \rho, \sigma\rangle \Downarrow \langle \mathsf{ret}\ v, \sigma'\rangle}\ \text{B-Ret}$$

$$\frac{\begin{array}{c}\mathrm{funcs}(p) = \texttt{Func}\ F_j(\overline{x_j : \tau_j}) : \tau'_j = b_j \quad (1 \le j \le k) \qquad \mathrm{stmts}(p) = \overline{s} \\ \ell_1, \ldots, \ell_k \text{ distinct} \qquad \rho_g = [F_1 \mapsto \ell_1, \ldots, F_k \mapsto \ell_k] \\ \sigma_0 = [\ell_1 \mapsto \mathsf{clo}(\overline{x_1},\ b_1,\ \rho_g), \ldots, \ell_k \mapsto \mathsf{clo}(\overline{x_k},\ b_k,\ \rho_g)]\end{array}}{}$$

3 Implementation

4 Distinctive-Feature Example Program

5 Reflection on AI and Collaboration

A Full Type System Rules

B Full Operational Semantics Rules